FloodSight

Methodology Note

Open-data flood intelligence and rainfall-triggered early-warning methodology for the Lagos pilot.

Prepared by	Rankine Innovation Lab
Pilot area	Eti-Osa, Lagos Island, and Kosofe, Lagos State
Version	1.0 - Prototype methodology

1. Purpose and scope

FloodSight is an open-data flood intelligence and early-warning project designed to help communities, planners, and emergency responders understand where flood risk is concentrated and when rainfall conditions should trigger early action. This methodology note documents how the Lagos pilot was produced and how the prototype can evolve into a dynamic operational application.

Current prototype output

The current FloodSight pilot produces a 30 m static flood-risk map, high-risk population exposure outputs, rainfall-triggered alert scenarios, real rainfall workflow tests, and evidence-ready maps and tables for communication and stakeholder review.

The methodology is intended to be transparent and reproducible. It explains the data sources, spatial processing, scoring logic, rainfall-triggered alert logic, output interpretation, limitations, and next development steps. The prototype should be interpreted as a flood susceptibility and alert-priority model, not a calibrated hydrodynamic flood-depth model.

Pilot geography: selected Lagos State local government areas - Eti-Osa, Lagos Island, and Kosofe. The pilot was selected because the area combines coastal exposure, dense development, waterways, low-lying terrain, and high population exposure.

Project status: working geospatial prototype with a clear pathway to automation, web deployment, and validation.

End-to-end methodology pipeline

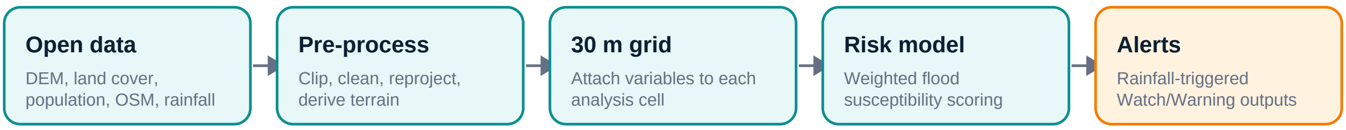

Figure 1. FloodSight converts open geospatial and rainfall data into static risk classes and rainfall-triggered alert outputs.

2. Data sources and project inputs

FloodSight uses only open or publicly accessible datasets. The prototype does not depend on proprietary datasets. Large raw datasets should remain outside the public repository and be referenced through their original public sources.

Theme	Dataset	Use in FloodSight
Administrative boundaries	Nigeria administrative boundary data	State and LGA boundaries for Lagos pilot selection.
Elevation	Copernicus DEM GLO-30	Base terrain layer used to derive elevation, slope, hillshade, filled DEM, and flow accumulation.
Land cover	ESA WorldCover	Land-cover class used to identify built-up, water, wetland, mangrove, vegetation, and other surface types.
Population	WorldPop	Population exposure estimate attached to the 30 m grid.
Roads and water bodies	OpenStreetMap	Context and exposure layers for roads, waterways, water bodies, and building-related features where available.
Rainfall	CHIRPS daily rainfall; future GPM IMERG	Prototype rainfall sampling and alert logic; future near-real-time update stream.

Data management approach

- Raw data is stored separately from processed data to preserve original downloads.
- Processed datasets are clipped to the Lagos pilot area and reprojected into a metre-based coordinate system for analysis.
- GeoPackage is preferred for vector outputs because it is more robust than shapefile for complex or large layers.
- Large raw datasets such as DEM tiles, OSM PBF, rainfall rasters, and population rasters should not be uploaded to GitHub. Public outputs and methodology documents can be shared instead.

3. Spatial framework

The modelling framework is based on a 30 m polygon grid covering the Lagos pilot area. Each grid cell is treated as an analysis unit. This creates a consistent spatial structure that allows FloodSight to attach terrain, land cover, rainfall, and exposure variables to comparable units.

Coordinate reference system

The working coordinate system is WGS 84 / UTM Zone 31N, commonly represented as EPSG:32631. This projected CRS is used because distances, grid spacing, slope-related operations, and proximity analysis require metre-based coordinates. Raw datasets in latitude/longitude are reprojected before distance-based analysis.

Grid creation

- Extract the Lagos pilot LGAs from the national administrative boundary dataset.
- Reproject the pilot boundary to EPSG:32631.
- Dissolve the three pilot LGAs into one pilot boundary for clipping and grid creation.
- Create a 30 m rectangle polygon grid using the pilot boundary extent.
- Clip the grid to the dissolved pilot area.

- Add cell_id, x_coord, and y_coord fields for indexing, joining, and downstream processing.

Why a grid?

A grid allows FloodSight to convert varied raster and vector data into one consistent analytical unit. This makes it easier to calculate flood score, population exposure, rainfall triggers, and dashboard summaries.

4. Terrain and environmental feature engineering

The DEM is the foundation of the terrain model. FloodSight derives several features from the elevation layer to capture topographic flood susceptibility.

Variable	Processing output	Role in model
Elevation	Mean elevation per grid cell	Lower elevation generally increases susceptibility to water accumulation and coastal/lagoon influence.
Slope	Slope derived from DEM	Flatter areas generally drain more slowly and can retain water.
Filled DEM	Sink-filled DEM	Used for hydrological conditioning before flow accumulation.
Flow accumulation	Flow concentration from filled DEM	Higher accumulation suggests places where runoff may concentrate.
Distance to water	Proximity raster from water bodies	Shorter distance to lagoons, rivers, canals, or water bodies increases susceptibility.
Land cover	Dominant land-cover class per cell	Built-up, wetland, mangrove, and water-adjacent classes are treated as higher-priority risk surfaces.
Population exposure	Population sum per cell	Used to estimate people potentially exposed in risk and alert zones.

OSM water bodies and distance-to-water

Water bodies are extracted from OpenStreetMap polygon features where possible. The extracted water layer is rasterized at 30 m resolution using the same extent as the modelling grid. A proximity raster is then created to calculate the distance from every grid cell to the nearest water feature. Where OSM line waterways are incomplete, polygon water bodies still provide a useful first approximation for distance-to-water susceptibility.

Rainfall handling

For the prototype, rainfall is handled in two ways. First, a manual stress-test rainfall scenario is used to verify that Watch/Warning logic works. Second, CHIRPS daily rainfall rasters are sampled at grid-cell centroids to support a real rainfall workflow. For operational deployment, GPM IMERG is recommended for near-real-time rainfall ingestion.

5. Flood-risk scoring method

The current FloodSight prototype uses a weighted overlay method. Each risk factor is normalised between 0 and 1, where 0 represents lower contribution to flood susceptibility and 1 represents higher contribution. The model is designed for transparent screening and prioritisation. It should be calibrated with observed flood events before being used as an official operational forecast.

Factor	Normalisation logic	Risk interpretation
Elevation	risk_elev = (max_elev - elev) / (max_elev - min_elev)	Lower terrain receives higher risk.
Slope	risk_slope = (max_slope - slope) / (max_slope - min_slope)	Flatter areas receive higher risk.
Flow accumulation	risk_flow = (flow - min_flow) / (max_flow - min_flow)	Higher contributing flow receives higher risk.
Distance to water	risk_waterdist = (max_dist - dist) / (max_dist - min_dist)	Closer to water receives higher risk.
Population	risk_pop = (pop - min_pop) / (max_pop - min_pop)	Used as exposure weight, not physical hazard.

Land-cover scoring

Land-cover classes are converted into risk contribution scores based on their relationship to flood susceptibility and exposure. Built-up surfaces, permanent water, wetland, mangrove, and sparse/bare surfaces are assigned higher scores than tree cover and lower-risk vegetation. These values should be refined with local drainage data and validation evidence in later phases.

Class	Land cover	Score	Interpretation
50	Built-up	1.0	High exposure and reduced infiltration context.
80	Permanent water	1.0	Water-adjacent or water-covered locations.
90	Herbaceous wetland	0.9	Wetland/flood-prone context.
95	Mangroves	0.8	Coastal/wetland interface.
60	Bare/sparse vegetation	0.6	Can increase runoff or exposure depending on context.
40	Cropland	0.5	Moderate surface sensitivity.
30	Grassland	0.4	Lower relative built-environment exposure.
20	Shrubland	0.3	Lower relative built-environment exposure.
10	Tree cover	0.2	Lower relative runoff/exposure contribution in this simple model.

6. Composite flood score and risk classes

The final flood score is computed as a weighted combination of the normalised factor scores. The weights reflect the first prototype assumptions and should be adjusted after validation with local flood observations, reports, satellite flood extents, or agency records.

Flood score formula

flood_score = 0.25 * risk_elev + 0.15 * risk_slope + 0.20 * risk_flow + 0.15 * risk_waterdist + 0.15 * risk_landcover + 0.10 * risk_pop

Factor	Weight	Reason
Elevation	0.25	Low-lying terrain is a key flood susceptibility factor.
Flow accumulation	0.20	Runoff concentration strongly influences flood-prone zones.
Slope	0.15	Flatter terrain can reduce drainage efficiency.
Distance to water	0.15	Proximity to water bodies increases exposure to overflow and coastal/lagoon influence.
Land cover	0.15	Built-up and wetland-related classes modify runoff and exposure.
Population	0.10	Captures impact/exposure priority rather than physical hazard alone.

Risk class	Score range	Interpretation
Low	0.00 - 0.24	Lower priority under current static susceptibility factors.
Moderate	0.25 - 0.49	Monitor and prepare where exposure is high.
High	0.50 - 0.74	Priority for preparedness, drainage checks, and planning.
Very High	0.75 - 1.00	Highest priority for early warning and emergency planning.

Risk interpretation

The risk classes should be interpreted as relative priority categories for the Lagos pilot. A Very High cell is not a guaranteed flood location in every rainfall event. Instead, it is a location where multiple susceptibility and exposure factors align and where preparedness, inspection, or early action may be more urgent.

7. Rainfall-triggered alert method

FloodSight separates the static flood-risk baseline from the dynamic rainfall-triggered alert layer. The static baseline changes slowly because elevation, distance to water, and land cover do not change every hour. The alert layer changes whenever new rainfall observations or forecasts are available.

Alert fields

Field	Meaning	Update behaviour
rain_24h	Recent or forecast rainfall over a 24-hour period, in millimetres.	Updated with new rainfall data.
rain_72h	Accumulated rainfall over a 72-hour period, in millimetres.	Updated with rolling rainfall accumulation.
alert_level	Current Watch/Warning/No Alert status for each grid cell.	Recalculated after rainfall update.

Prototype alert rules

Condition	Alert level
High or Very High risk and rain_24h >= 100 mm, or rain_72h >= 150 mm	Warning
High or Very High risk and rain_24h >= 50 mm, or rain_72h >= 100 mm	Watch
Moderate risk and rain_24h >= 100 mm, or rain_72h >= 150 mm	Watch
All other combinations	No Alert

Dynamic app principle

The risk map is the baseline. The alert map is dynamic. After deployment, the application will refresh rainfall feeds, update rain_24h and rain_72h, recalculate alert_level, and refresh the web dashboard automatically.

8. Outputs and interpretation

The prototype produces map and table outputs that communicate both hazard priority and exposure. The main outputs are intended for planners, emergency actors, community organisations, researchers, and potential deployment partners.

Output	Interpretation
Static flood-risk map	Shows Low, Moderate, High, and Very High risk zones based on terrain, land cover, water proximity, flow, and exposure variables.
High-risk population exposure map	Shows where population is concentrated within High and Very High risk areas.
Demo rainfall alert scenario map	Demonstrates how Watch/Warning zones are triggered under a simulated heavy rainfall event.
Real rainfall alert map	Shows alert status for selected real rainfall dates. No Alert is valid when rainfall is below threshold.
Population by risk class table	Summarises exposed population by risk class.
Population by alert level table	Summarises exposed population under scenario or real alert conditions.
Web map export	Provides a lightweight interactive viewer for toggling risk, exposure, alert, road, and water layers.

Recommended map sequence for communication

- Study area: show pilot boundary, roads, and water bodies.
- Flood-risk zones: show baseline susceptibility classes.
- Population exposure: show where people are concentrated inside high-risk areas.
- Demo alert scenario: show how heavy rainfall triggers Watch/Warning zones.
- Real rainfall alert: show the operational logic, including No Alert when rainfall is below threshold.

9. Quality assurance checks

The following checks are recommended before publishing or presenting FloodSight outputs.

Check	Expected result
CRS check	All processed analysis layers use EPSG:32631 or the chosen metre-based local projection.
Extent check	Pilot layers align over Eti-Osa, Lagos Island, and Kosofe and do not shift away from Lagos.
Grid check	30 m grid is clipped to the pilot boundary and contains unique cell_id values.
Raster check	DEM, slope, flow, land cover, population, distance-to-water, and rainfall layers overlap the grid.
Attribute check	Scored grid contains risk_class, flood_score, pop_sum, rain_24h, rain_72h, and alert_level where relevant.
Visual check	Risk classes display with no dense black grid outlines; dissolved display layers are used for public maps.
Scenario check	Rainfall stress-test produces Watch/Warning zones when thresholds are exceeded.

10. Limitations

FloodSight is a strong early pilot, but its outputs should be interpreted within the limits of available data and the current modelling approach.

- The current model is a flood susceptibility and alert-priority model, not a calibrated hydrodynamic simulation.
- Flood depth is not yet predicted. The current output indicates where risk and exposure are higher, not how deep water will be.
- The scoring weights are expert-rule assumptions for the prototype and should be calibrated with observed flood events.
- CHIRPS rainfall is useful for daily historical and monitoring workflows but is coarse compared with a 30 m grid. For operational alerts, GPM IMERG or other near-real-time rainfall feeds should be integrated.
- OpenStreetMap coverage may be incomplete for drainage canals, small waterways, and building details.
- Population estimates are modelled gridded data and should be treated as exposure approximations.
- Local drainage capacity, blocked drains, tides, storm surge, culvert conditions, and sewer backflow are not yet fully represented.

Flood-depth prediction

Flood depth should be added in a later development phase. The current prototype does not claim to estimate exact water depth. A future depth module may begin as a depth-proxy model using elevation, rainfall, flow accumulation, water proximity, land cover, and validation records. A more rigorous version would use hydrodynamic modelling or machine learning trained with observed flood depths or satellite-derived flood extents.

11. Full application pathway

To become a full working FloodSight application, the QGIS-based prototype should evolve into an automated geospatial data platform.

Component	Full app requirement
Data pipeline	Automated ingestion of rainfall, forecast, exposure, and updated open geospatial data.
Database	PostgreSQL/PostGIS storage for grid cells, geometries, risk scores, rainfall values, alerts, and summaries.
Risk engine	Python or backend service to recalculate flood score, rainfall triggers, and exposure summaries.
Dashboard	Web map interface for risk zones, alert levels, exposed population, reports, and downloads.
Notifications	Email, SMS, WhatsApp, or API alerts for Watch/Warning conditions.
Validation	Comparison with flood reports, historical events, satellite flood extents, and stakeholder feedback.
Depth module	Future flood-depth proxy, machine-learning, or hydrodynamic model after validation data is available.

Suggested deployment architecture

- Frontend: web dashboard using Leaflet, MapLibre, React, or a lightweight static web map for the first deployment.
- Backend: FastAPI, Django, or Node.js API for data services and alert logic.
- Database: PostGIS for spatial querying and tiled map generation.
- Scheduler: cron, Celery, GitHub Actions, or cloud scheduler for regular rainfall updates.

- Hosting: low-cost cloud VM, managed database, or serverless architecture depending on scale.
- Repository: public GitHub repository for methodology, output maps, tables, and reproducible code; raw large datasets remain referenced externally.

12. Next development milestones

Period	Milestone
0-1 month	Clean repository, publish methodology, complete web map demo, organise data documentation.
1-2 months	Automate rainfall ingestion and generate rolling 24 h and 72 h rainfall summaries.
2-4 months	Build lightweight dashboard, expose map layers, and add downloadable reports.
4-6 months	Validate against historical flood events, community reports, and satellite flood evidence.
6-9 months	Add road/building exposure, LGA-level summaries, and notification workflows.
9-12 months	Pilot with stakeholders, refine thresholds, expand to additional Lagos LGAs and other Nigerian cities.

Summary

FloodSight is already a functioning geospatial prototype: it creates flood-risk zones, population exposure layers, and rainfall-triggered alert scenarios. The next step is to automate data ingestion, deploy the dashboard, validate the model, and expand from risk and alert mapping toward flood extent and flood-depth intelligence.

Appendix: recommended public repository contents

A public repository should include README.md, this methodology PDF, exported maps, summary tables, and optional web demo files. It should not include raw DEM, CHIRPS rasters, WorldPop rasters, OSM PBF files, or large GeoPackage processing files.